

**B.Sc in Computer Science and Engineering Thesis  
Report on**

**Poly-Dialectal Neural Machine  
Translation System for Bangla  
Regional Dialects**

**Submitted By**

**Rakib Ullah**

Reg No: 2020331520

**Ruhul Islam Rahul**

Reg No: 2020331529

**Tanbir Ahmed**

Reg No: 2020331516

**Supervised By**

**Nayan Kumar Nath**

Lecturer

Department of Computer Science and Engineering,  
Sylhet Engineering College

**Department of Computer Science and Engineering  
SYLHET ENGINEERING COLLEGE,  
SYLHET**

Affiliated with  
**Shahjalal University of Science and Technology**  
Sylhet, Bangladesh.

22 August, 2026

# 1 Introduction

## 1.1 Background and Motivation

Bangla (Bengali), an Indo-Aryan language, stands as one of the most widely spoken languages globally, boasting over 240 million native speakers primarily distributed across Bangladesh and the Indian states of West Bengal, Tripura, and Assam. Despite its immense socio-linguistic prominence and rich literary heritage, Bangla is not a monolithic entity; rather, it exhibits substantial intra-language diversity through a vast array of distinct regional dialects. Dialects such as Sylheti, Chittagonian, Barisali, Noakhali, Rangpuri, Rajshahi, and Mymensingh differ markedly from Standard Colloquial Bangla (SCB) across multiple linguistic strata. These divergences manifest as profound phonological variations, complex morphological inflections, altered syntactic structures, and highly localized lexical choices. In extreme cases, such as the Chittagonian or Sylheti dialects, the linguistic deviation is so severe that they are often bordering on mutual unintelligibility with Standard Bangla.

This deep intra-language divergence engenders significant communication barriers within the broader socio-economic framework. Native speakers of peripheral regional dialects frequently encounter difficulties in seamless communication with speakers of other dialects, as well as in formal interactions, legal proceedings, and educational instructions traditionally conducted in Standard Bangla. Furthermore, modern Natural Language Processing (NLP) tools, machine translation systems, and digital interfaces—ranging from virtual assistants to automated governance platforms—are overwhelmingly optimized exclusively for Standard Bangla. Consequently, millions of dialectal speakers face systemic digital marginalization and language-access inequality, lacking automated communication technologies tuned to their native spoken forms. This technological disparity highlights an urgent need for inclusive NLP systems that can bridge the communication gap without enforcing linguistic homogenization.

## 1.2 Problem Statement

In recent years, Neural Machine Translation (NMT) architectures and Large Language Models (LLMs) have achieved state-of-the-art results for cross-lingual tasks involving high-resource languages. However, their efficacy degrades severely when applied to low-resource intra-language dialectal shifts. Most existing Bangla NLP benchmarks and pre-trained models treat the language as a singular, uniform distribution, fundamentally ignoring the complex cross-dialectal matrix.

This oversight stems from a confluence of critical technical challenges. Firstly, there is an acute scarcity of multi-dialectal parallel corpora required to effectively fine-tune deep neural networks. Secondly, the lack of standardized orthography for regional dialects exacerbates tokenization mismatches; standard Byte-Pair Encoding (BPE) [17] or Word-Piece [24] algorithms optimized for SCB frequently fragment dialectal words into meaningless subwords, destroying semantic integrity. Finally, existing regional text datasets suffer from severe class imbalance and are predominantly tailored for simple classification tasks rather than complex generative alignment. Addressing these challenges requires a paradigm shift from uni-dialectal modeling to a holistic, poly-dialectal approach.

### 1.3 Proposed System and Contributions

To address these fundamental limitations, this research introduces a unified **Poly-Dialectal Neural Machine Translation System** tailored specifically for the morphological and syntactic nuances of Bangla regional dialects. As illustrated in Figure 1, our architecture enables robust, multi-directional translation across three core operational paradigms: Dialect  $\rightarrow$  Standard Bangla, Standard Bangla  $\rightarrow$  Dialect, and direct Dialect  $\rightarrow$  Dialect translation, thereby eliminating the cascading errors typically introduced by pivot-based translation methods.

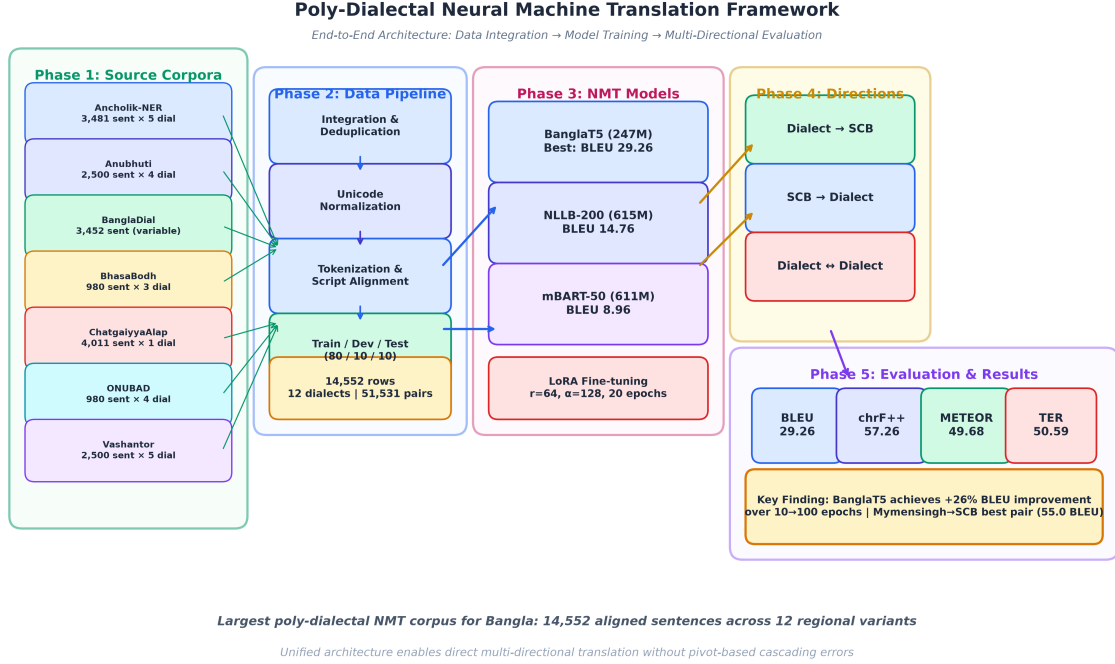


Figure 1: End-to-end architectural workflow of the Poly-Dialectal Neural Machine Translation Framework, depicting five phases: source corpora integration from seven existing datasets, the data processing pipeline, NMT model fine-tuning with LoRA, multi-directional translation paradigm, and evaluation with four standard MT metrics.

The primary contributions of this thesis are summarized as follows:

1. **Largest Multi-Dialectal Parallel Corpus:** We construct the most expansive multi-dialect machine translation corpus for Bangla to date, comprising 14,552 aligned rows across 12 regional dialect columns. The corpus integrates seven prior datasets—Ancholik-NER, Anubhuti, BanglaDial, BhasaBodh, ChatgaiyyaAlap, ONUBAD, and Vashantor. Additionally, we incorporate 500 manually created, bidirectional, and verified high-quality parallel sentence pairs for five dialects (Rangpur, Tangail, Kishoreganj, Narail, Narsingdi), yielding over 51,531 non-null parallel pairs across 11 regional variants.
2. **Pioneering Benchmark of NMT Architectures on Poly-Dialectal Translation:** To the best of our knowledge, this work presents the first systematic empirical evaluation of prominent sequence-to-sequence architectures—specifically BanglaT5 [2], NLLB-200 [22], and mBART-50 [21]—on the novel task of multi-directional, poly-dialectal Bangla translation.

3. **Systematic Cross-Dialectal Linguistic Analysis & Scaling Study:** We conduct a rigorous Optimal Dataset Size Study (progressively scaling from 5,000 instances) and quantitative characterization of cross-dialectal transfer mechanisms. We uncover how morphological proximity and training data volume jointly influence neural machine translation efficacy across diverse metrics.
4. **Real-World Web Deployment:** We deploy the best-performing NMT model in a fully functional web application, bridging the digital language divide by offering real-time poly-dialectal translation capabilities accessible to end users.

## 1.4 Thesis Organization

The remainder of this report is organized as follows: Section 2 critically reviews related literature on Bangla regional NLP and dialectal machine translation. Section 3 formally states the research hypothesis and objectives. Section 4 details the methodology, including dataset construction, alignment strategies, and model configurations. Section 5 presents a comprehensive evaluation of the experimental results. Section 6 discusses the limitations of the current framework and proposes future research directions, followed by concluding remarks in Section 7.

## 2 Literature Review

### 2.1 Overview of Regional Bangla NLP

Historically, the development of linguistic resources for Bangla has heavily favored Standard Colloquial Bangla, leaving regional dialects severely underrepresented in the NLP ecosystem. However, over the past few years, interest in low-resource regional Bangla NLP has seen a significant upsurge. Initial studies focused predominantly on resource creation for text classification and sentiment analysis tasks for specific regions.

Sultana et al. [20] established the *BdRegionText* corpus, a pioneering effort for regional text classification, which starkly demonstrated the performance bottleneck of standard classifiers when exposed to rich dialectal vocabulary. Building on the need for broader representation, Mahi et al. [11] released *BanglaDial*, a merged text dataset that brought to light the severe class imbalance issues and data scarcity prevalent across different regional dialects.

In specialized downstream NLP tasks, the complexity of dialectal morphology continues to pose significant hurdles. Kundu et al. [9] introduced *ANUBHUTI*, a dedicated corpus targeting sentiment analysis in regional languages, revealing that emotion markers vary significantly across regional boundaries. For the crucial task of offensive speech identification, Fayaz et al. [6] developed *BIDWESH*, evaluating hate speech detection mechanisms across regional dialects and proving that SCB-trained toxicity classifiers fail to generalize to dialectal slang. Concurrently, Paul et al. [14] formulated *ANCHOLIK-NER*, providing essential named entity annotations across regional variants to facilitate information extraction in non-standard texts.

### 2.2 Dialectal Machine Translation and LLM Benchmarks

While text classification laid the groundwork, direct efforts in generative machine translation for Bangla dialects have emerged only recently, driven by the advent of massively multilingual neural models. Early translation corpora were highly localized. For instance, Chowdhury et al. [4] built *ChatgaiyyaAlap*, a dedicated parallel corpus strictly designed for translating Chittagonian dialect sentences to Standard Bangla. To expand the dialectal scope, Sultana et al. [19] presented *ONUBAD*, providing an initial generalized resource for automated regional dialect conversion. This was followed by Faria et al. [5], who introduced *Vashantor*, a multilingual benchmark covering five regional dialects. Though a significant step forward, *Vashantor* remained constrained by its limited volume, offering only 2,500 sentences per dialect.

More recently, the research community has pivoted towards exploring the zero-shot and few-shot capabilities of Large Language Models (LLMs) for dialect translation. Mahjabin et al. [12] constructed *BanglaCHQ-Prantik* to benchmark state-of-the-art LLMs against human-evaluated dialect translation standards. Similarly, Jawad et al. [8] presented *DIALTSA-BN*, evaluating LLM capabilities in joint dialect translation and sentiment analysis. Furthermore, Bhuiyan et al. [3] explored *BhasaBodh*, investigating methods to bridge Bangla dialects and romanized forms through advanced machine translation techniques. Despite these advancements, fine-tuned NMT models consistently outperform generalized LLMs in specialized low-resource dialectal translation due to the LLMs' inherent lack of dialect-specific pre-training data.

## 2.3 Critical Synthesis and Research Gap

While the aforementioned contributions have collectively laid a valuable foundation for Bangla dialectal NLP, a critical review reveals several major structural and methodological shortcomings that currently stifle the development of robust, production-ready translation systems:

1. **Single-Dialect Isolation:** Corpora such as *ChatgaiyyaAlap* focus exclusively on a single dialect pair (e.g., Chittagonian-Standard). Models trained on such isolated datasets fail entirely to generalize across other regional dialects, necessitating the training of separate, disjoint models for every region.
2. **Severe Corpus Volume Limitations & Lack of Scaling Studies:** Multi-dialect datasets like *Vashantor* offer only modest data volumes (e.g., 2,500 samples per dialect). This scale is vastly insufficient for deep fine-tuning of architectures like mT5 or mBART-50. Furthermore, prior literature entirely lacks systematic scaling studies to empirically determine optimal dataset size thresholds for cross-dialectal transfer.
3. **The Pivot Translation Problem:** Existing works lack a unified framework capable of translating directly between different regional dialects (e.g., Sylheti  $\rightarrow$  Chittagonian). Current systems force a “pivot” approach, routing translation through Standard Bangla as an intermediary, which compounds translation errors and degrades final semantic accuracy.
4. **Absence of Real-World Deployment:** State-of-the-art models developed in prior studies remain confined to offline evaluation environments. The community lacks open, deployed web applications that can provide real-time translation services directly to marginalized dialect speakers.

To definitively resolve these limitations, our work introduces a comprehensive, balanced, poly-dialectal parallel corpus that is orders of magnitude larger than previous efforts. By fine-tuning state-of-the-art Transformer models on this expansive dataset, we enable direct, multi-directional, poly-dialectal translation within a single, unified neural architecture.

## 2.4 Summary of Dialectal Resources

Table 1 summarizes the major dialectal datasets and resources reviewed in this section, detailing their coverage, architecture, evaluation metrics, and state-of-the-art performance.

Table 1: Comprehensive Summary of Bangla Regional Dialect NLP Resources and Datasets

| Dataset & Year                            | Primary Task & Domain                                                                 | Dialects & Dataset Scale                                                                          | Methodology & Quality Control                                                                                                           | Key Metrics & SOTA Findings                                                                                                                       |
|-------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. ANCHOLIK-<br>NER<br>(2025)             | Dialectal<br>Named<br>Entity<br>Recognition<br>(NER); 10<br>entity classes.           | Sylhet,<br>Chittagong,<br>Barishal.<br>10,443<br>sentences<br>(3,481/di-<br>alect).               | BIO tagging.<br>Sourced from<br>Vashantor<br>(71.8%),<br>ONUBAD, &<br>news. 3-annotator<br>consensus pipeline.                          | First dialectal NER<br>baseline. Assamese NER<br>baseline reached 80.69%<br>F1 (MuRIL); medical<br>BanglishBERT got<br>56.13% F1.                 |
| 2. ANUB-<br>HUTI<br>(2026)                | Thematic<br>Sentiment &<br>Multilabel<br>Emotion<br>Analysis in<br>dialectal<br>text. | Mymensingh,<br>Noakhali,<br>Sylhet,<br>Chittagong.<br>10,000<br>sentences<br>(2,500/re-<br>gion). | Sourced from<br>MONOVAB. 8<br>native translators<br>(2/region) with<br>4-stage QA<br>(sentiment, NaN<br>parsing, anomaly,<br>spelling). | High inter-rater<br>agreement (Cohen’s<br>$\kappa = 0.76\text{--}0.84$ ).<br>Dominant emotions:<br>Contempt (921), Disgust<br>(871), Anger (732). |
| 3. BIDWESH<br>(2025)                      | Multi-<br>Dialectal<br>Hate Speech<br>& Toxicity<br>Detection.                        | Barishal,<br>Noakhali,<br>Chittagong.<br>9,183<br>instances<br>(3,061 paral-<br>lel/dialect).     | Sourced from<br>BD-SHS;<br>translated by 6<br>native speakers;<br>5-stage validation.<br>Target/Type<br>taxonomy.                       | Balanced binary split<br>(49.2% Hate / 50.8%<br>Non-Hate). Individual<br>targets highest (34.4%);<br>Slander dominant<br>(53.7%).                 |
| 4. Bangla<br>Dialect<br>Bibliogra-<br>phy | Systematic<br>Bibliography<br>&<br>Meta-Review<br>of Bangla<br>Dialect NLP.           | Comprehensive<br>overview<br>across major<br>regional<br>variants in<br>BD.                       | Systematic<br>synthesis tracking<br>11 publications,<br>annotation<br>strategies, &<br>baselines.                                       | Mapped 11 unique<br>corpora; highlights<br>Sylheti & Chittagonian<br>as most studied;<br>Barishal, Mymensingh,<br>Noakhali emerging.              |
| 5. BanglaDial<br>(2025)                   | Dialect<br>Identification<br>(DID) &<br>Class<br>Imbalance<br>Modeling.               | 11 Dialects +<br>Standard<br>Bangla.<br>60,729<br>sentences<br>(362,430<br>words).                | Aggregated from 4<br>corpora<br>(Vashantor,<br>ONUBAD, etc.).<br>Deduplicated<br>(4.1% removed) &<br>normalized.                        | Long-tail distribution<br>(Top-3: 41.3% of<br>records). Imbalance<br>Ratio $IR = 9.82$ ;<br>Shannon Entropy<br>$H = 3.416$ bits.                  |

Continued on next page



Table 1 – continued from previous page

| Dataset & Year              | Primary Task & Domain                                                       | Dialects & Dataset Scale                                                              | Methodology & Quality Control                                                                                         | Key Metrics & SOTA Findings                                                                                                         |
|-----------------------------|-----------------------------------------------------------------------------|---------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| 6. BdRe-gionText (2022)     | Bangla Regional Text Classification on social media content.                | Chittagong, Noakhali, Barishal, Rangpur. 2,573 sample texts.                          | Scraped from FB/YouTube/IG with human validation. Extracted TF-IDF, CountVectorizer, & BoW.                           | Random Forest + TF-IDF achieved peak 79.15% accuracy (10-fold CV: 79.47%, weighted F1: 0.78). RF + BoW got 75.92%.                  |
| 7. BhasaBodh (2025)         | Script Normalization & Multilingual Dialect Translation (Native/Romanized). | Chittagong, Sylhet (aligned with Standard Bangla & English). 1,960 sentences.         | Sourced from ONUBAD; augmented via Gemini 2.5 Pro for Romanizations. Fine-tuned NLLB-200 & mBART-50.                  | mBART-50 achieved peak 87.44 BLEU on Romanized → Standard Bangla and 74.36 BLEU on Chittagong ↔ Sylhet.                             |
| 8. Chat-gaiyyaAlap (2025)   | Dialect Conversion (Chittagonian → Standard Bangla).                        | Chittagonian (aligned with Standard Bangla). 4,012 sentences + 1,500-word dictionary. | Sourced from social media/dramas; validated by 5 native Chittagonian graduates.                                       | Created clean translation mapping resolving severe grammatical divergences & phonetic shifts in negative sentences.                 |
| 9. BanglaCHQ-Prantik (2025) | Medical-Domain Machine Translation (Standard Bangla → Dialects).            | Sylheti (2,350 sentences), Chittagonian (500 sentences).                              | Sourced from BanglaCHQ-Summ; validated by 11 validators & 2 physicians. Evaluated LLMs (Qwen, Gemma, GPT-4o, Gemini). | Gemini 2.5 Flash achieved SOTA on Sylheti (23.67 BLEU); GPT-4o with CoT achieved SOTA on Chittagonian (24.67 BLEU).                 |
| 10. DIALTSA-BN (2025)       | Dialect Translation & Sentiment Analysis (Native vs. Transliterated).       | Chattogram, Barishal, Sylhet, Noakhali. 600 annotated instances (150/region).         | Sourced from YouTube; filtered via Word2Vec. Evaluated Claude 3.7, GPT-4o-mini, Gemma-3, Llama-3.1, etc.              | Transliteration boosted translation BLEU from 0.07 to 0.330 (Claude 3.7) & 0.317 (GPT-4o-mini). Few-shot sentiment F1 reached 0.98. |

Continued on next page

Table 1 – continued from previous page

| Dataset & Year                                 | Primary Task & Domain                                                        | Dialects & Dataset Scale                                                                        | Methodology & Quality Control                                                                                                              | Key Metrics & SOTA Findings                                                                                                                        |
|------------------------------------------------|------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| 11.<br>BanglaCHQ-Prantik<br>Baseline<br>(2025) | Pediatric & Adult<br>Clinical MT<br>Error<br>Taxonomy.                       | Sylheti,<br>Chittagonian.<br>2,850 total<br>aligned<br>queries.                                 | Parallel clinical<br>reference tracking.<br>Highlighted<br>phonetic shifts<br>(e.g., *pete* →<br>*feto* / *cano*).                         | Identified 4 main LLM<br>failure modes:<br>terminology mismatch,<br>dosage corruption<br>("500mg" → "5 gram"),<br>idioms, & orthographic<br>drift. |
| 12.<br>ONUBAD<br>(2025)                        | Fine-Grained<br>Dialect<br>Conversion &<br>Neural<br>Machine<br>Translation. | Chittagong,<br>Sylhet,<br>Barisal. 7,950<br>total parallel<br>rows (980 sen-<br>tences/region). | Sourced from liter-<br>ature/volunteers;<br>validated by<br>university<br>linguists.<br>Benchmarked<br>Seq2Seq &<br>Transformers.          | Documented 2<br>words/phrase & 3<br>words/sentence.<br>BanglaT5 and<br>SeamlessM4T<br>established peak NMT<br>baselines.                           |
| 13.<br>Vashantor<br>(2023)                     | Multi-Task<br>Dialectal<br>NMT &<br>Region Clas-<br>sification.              | Chittagong,<br>Noakhali,<br>Sylhet,<br>Barishal,<br>Mymensingh.<br>32,500 total<br>sentences.   | Sourced from<br>social media/news;<br>validated with<br>Cohen's & Fleiss'<br>Kappa. Custom<br>DialectBanglaT5<br>& Dialect-<br>BanglaBERT. | DialectBanglaT5<br>achieved SOTA on<br>Mymensingh (71.93<br>BLEU).<br>DialectBanglaBERT<br>reached 89.02% region<br>classification accuracy.       |

## 3 Research Hypothesis and Objectives

### 3.1 Research Hypothesis

We posit the following hypothesis: *A neural machine translation model trained on a unified, multi-dialectal parallel corpus encompassing multiple Bangla regional variants will demonstrate significantly improved generalization and cross-dialectal transfer compared to models trained on isolated, uni-dialectal datasets.* Specifically, we hypothesize that a poly-dialectal training paradigm exploits shared morphological and syntactic structures across related dialects, enabling the model to learn transferable representations that benefit even low-resource dialect pairs.

### 3.2 Research Objectives

The specific objectives of this research are:

1. To construct the largest multi-dialect parallel corpus for Bangla machine translation by integrating, aligning, and deduplicating seven existing dialectal datasets across 12 regional variants.
2. To develop a unified poly-dialectal translation system capable of three translation paradigms: Dialect  $\rightarrow$  Standard Bangla, Standard Bangla  $\rightarrow$  Dialect, and direct Dialect  $\rightarrow$  Dialect translation.
3. To systematically evaluate and compare the performance of three state-of-the-art multilingual NMT architectures—BanglaT5 [2], NLLB-200 [22], and mBART-50 [21]—under parameter-efficient LoRA [7] fine-tuning on the constructed corpus.
4. To quantitatively analyze cross-dialectal transfer effects by examining how training data volume and linguistic proximity jointly influence translation quality.
5. To promote digital inclusion and linguistic accessibility for Bangla dialect speakers by establishing open benchmarks for future research.

## 4 Methodology

### 4.1 Dataset Construction and Integration

The foundation of our poly-dialectal NMT system is a meticulously curated multi-dialect parallel corpus. We integrate seven publicly available dialectal datasets, each contributing varying levels of coverage across different regional variants:

Table 2: Source corpus contributions per dialect. Values indicate the number of parallel sentence pairs available from each source.

| Source Corpus  | SCB           | Syl          | Cht           | Bar          | Mym          | Noa          | Ran          | Raj        |
|----------------|---------------|--------------|---------------|--------------|--------------|--------------|--------------|------------|
| Ancholik-NER   | 3,481         | 3,481        | 3,481         | 3,481        | 3,481        | 3,481        | 0            | 0          |
| Anubhuti       | 2,500         | 2,500        | 2,500         | 0            | 2,500        | 0            | 0            | 0          |
| BanglaDial     | 3,452         | 442          | 577           | 790          | 712          | 0            | 655          | 891        |
| BhasaBodh      | 980           | 980          | 980           | 0            | 0            | 0            | 0            | 0          |
| ChatgaiyyaAlap | 4,011         | 0            | 4,011         | 0            | 0            | 0            | 0            | 0          |
| ONUBAD         | 980           | 980          | 980           | 980          | 0            | 0            | 0            | 0          |
| Vashantor      | 2,500         | 2,500        | 2,500         | 0            | 2,500        | 2,500        | 0            | 0          |
| <b>Total</b>   | <b>13,556</b> | <b>6,422</b> | <b>10,567</b> | <b>4,270</b> | <b>5,712</b> | <b>5,000</b> | <b>1,155</b> | <b>891</b> |

The final integrated dataset comprises 14,552 rows across 12 dialect columns (Standard Bangla plus 11 regional variants: Sylheti, Chittagonian, Barisali, Mymensingh, Noakhali, Rangpuri, Rajshahi, Kishoreganj, Narail, Narsingdi, and Tangail), yielding 51,531 total non-null parallel sentence pairs. This represents the largest such corpus for Bangla to date. Crucially, to mitigate the extreme scarcity of data for certain regional variants, we manually created a high-quality dataset consisting of altogether 2,500 bidirectional parallel sentence pairs (Standard Bangla  $\leftrightarrow$  Dialect) distributed evenly across five dialects: Rangpur, Tangail, Kishoreganj, Narail, and Narsingdi (500 pairs each). To ensure rigorous data quality, each of these dialectal subsets was independently evaluated and verified by three native speakers of the corresponding dialect.

As depicted comprehensively in Figure 2, despite augmentation efforts, the dataset exhibits significant class imbalance—Chittagonian commands 10,567 entries while the newly added dialects (e.g., Narail, Tangail) are constrained to approximately 500 pairs. This unified visualization delineates both the gross dialectal volume and the exact decomposition of contributing source corpora for each variant. Furthermore, Figure 3 presents the cross-dialect parallel coverage matrix, quantitatively illustrating the pairwise intersection of translated sentences across all regional variants.

### 4.2 Data Preprocessing Pipeline

The raw integrated corpus undergoes a rigorous four-stage preprocessing pipeline:

1. **Unicode Normalization:** All text is normalized to Unicode NFC form to resolve inconsistent character compositions prevalent in Bangla script.
2. **Cleaning and Deduplication:** Duplicate entries across source datasets are identified and removed. Empty, whitespace-only, and non-Bangla script entries are filtered.

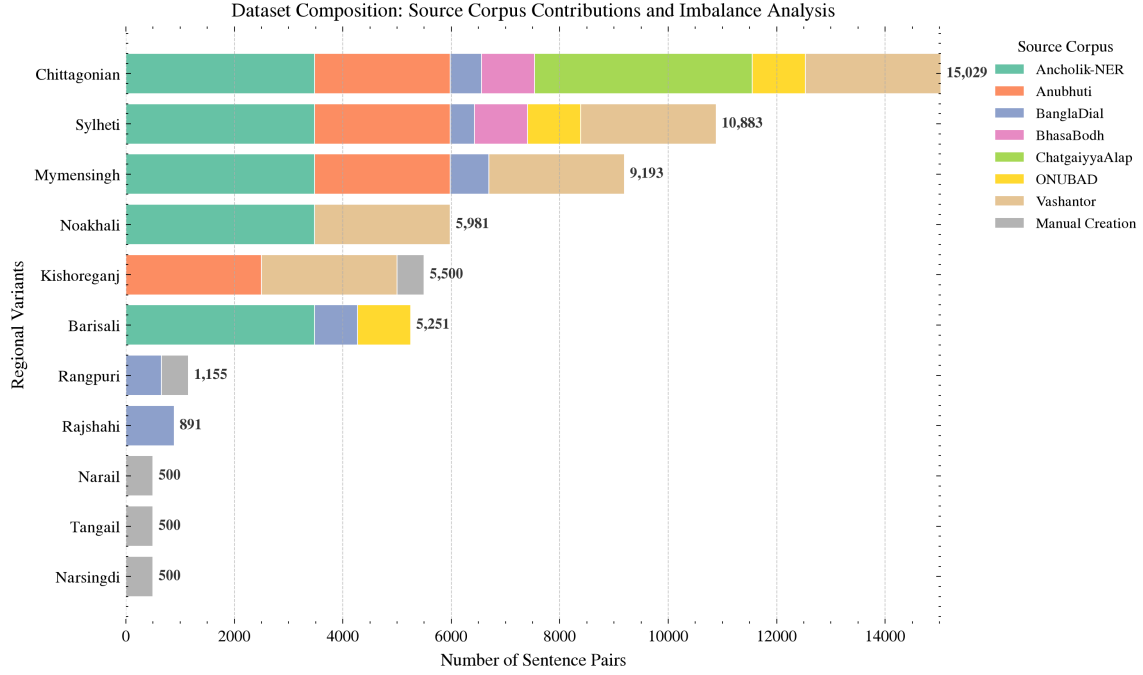


Figure 2: Dataset Composition and Source Provenance: A stacked horizontal distribution illustrating the severe class imbalance alongside the precise proportional contributions from prior corpora and manual augmentation efforts for each regional dialect.

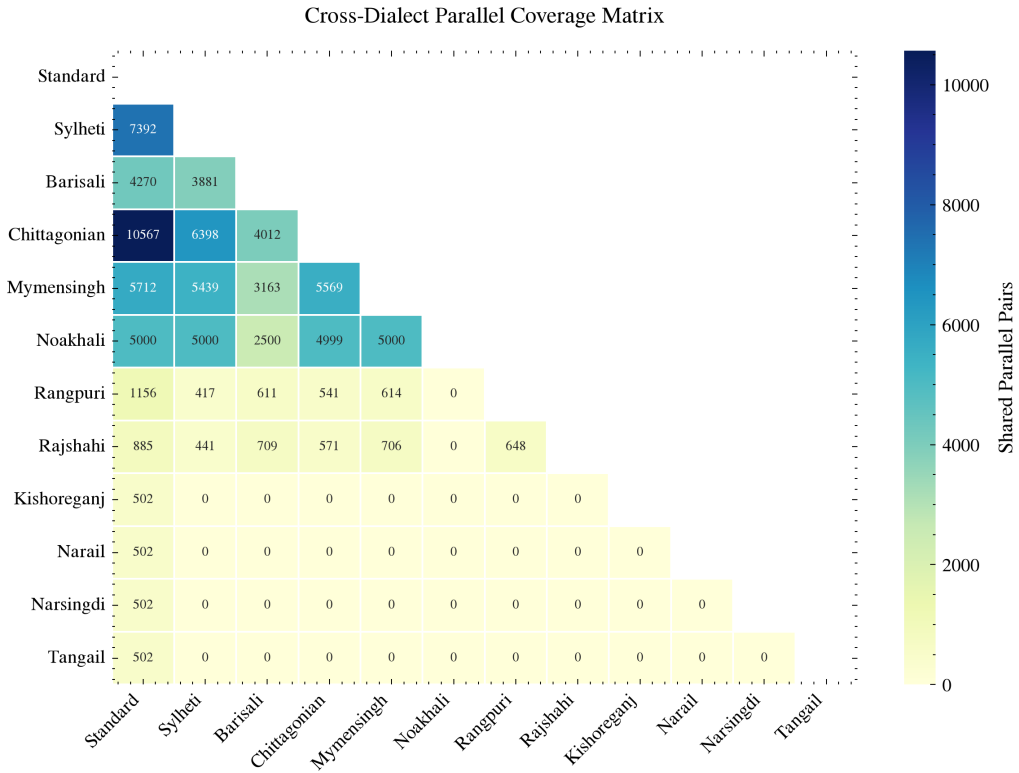


Figure 3: Cross-Dialect Parallel Coverage Matrix: An upper-triangular masked heatmap detailing the exact density of shared, parallel sentence pairs intersecting any two evaluated dialectal variants.

3. **Tokenization and Script Alignment:** Text is tokenized using model-specific SentencePiece tokenizers. For BanglaT5, the 32K vocabulary is optimized for Bangla NLG tasks; for NLLB-200 and mBART-50, the larger vocabularies ( $\sim 256K$  and  $\sim 250K$ , respectively) accommodate multilingual coverage.
4. **Train/Dev/Test Split:** The corpus is partitioned into training (80%), development (10%), and test (10%) sets with stratified sampling to ensure proportional dialect representation across splits.

### 4.3 Model Architectures

We evaluate three state-of-the-art encoder-decoder Transformer [23] architectures, chosen to represent a spectrum of design philosophies for multilingual and Bangla-specific NMT. The architectural comparison is summarized in Table 3.

Table 3: Architectural comparison of the three NMT models evaluated in this study.

| Feature                | BanglaT5 | NLLB-200    | mBART-50    |
|------------------------|----------|-------------|-------------|
| Parameters             | 247M     | 615M        | 611M        |
| Architecture           | T5       | Transformer | BART        |
| Encoder/Decoder Layers | 12/12    | 12/12       | 12/12       |
| Hidden Dimension       | 768      | 1,024       | 1,024       |
| Attention Heads        | 12       | 16          | 16          |
| FFN Dimension          | 3,072    | 4,096       | 4,096       |
| Vocabulary Size        | 32K      | $\sim 256K$ | $\sim 250K$ |
| Languages Supported    | Bangla   | 200         | 50          |

#### 4.3.1 BanglaT5

**Overview and Capabilities:** BanglaT5 [2] is a sequence-to-sequence model based on the T5 [16] architecture, pre-trained extensively on approximately 27.5 GB of monolingual Bangla text. **Advantages and Limitations:** Being a language-specific model, its 32K vocabulary is heavily optimized for Bangla morphology, preventing the excessive subword fragmentation commonly seen in multilingual tokenizers. However, it lacks cross-lingual pre-training signals that might otherwise aid in zero-shot generalization. **Suitability for Dialects:** We selected BanglaT5 because its deep grounding in Standard Bangla provides a strong prior for generating morphologically complex dialectal variations. It serves as the primary language-centric baseline in our study.

#### 4.3.2 NLLB-200

**Overview and Capabilities:** NLLB-200 [22] (No Language Left Behind) is Meta AI’s massively multilingual Transformer designed for direct translation across 200 languages. We utilize the distilled 600M parameter variant. **Advantages and Limitations:** Its primary advantage is robust multilingual alignment and a vast  $\sim 256K$  vocabulary designed to encompass diverse scripts. However, this large vocabulary dilutes its representational capacity for any single low-resource language, potentially hampering its nuance in intra-language dialectal shifts. **Suitability for Dialects:** NLLB-200 is selected to determine if

massive multilingual pre-training provides implicit transfer learning benefits for regional dialects.

### 4.3.3 mBART-50

**Overview and Capabilities:** mBART-50 [21] is a multilingual denoising auto-encoder based on the BART [10] architecture, extended to cover 50 languages. **Advantages and Limitations:** It excels at sequence generation tasks due to its denoising objective during pre-training. Like NLLB-200, its limitation lies in the generalized multilingual tokenization which is not exclusively tuned for Bangla scripts. **Suitability for Dialects:** We include mBART-50 as a strong baseline for multilingual generation to compare against both the language-specific BanglaT5 and the massively multilingual NLLB-200.

## 4.4 Hyperparameters

To ensure rigorous reproducibility and optimal convergence, Table 4 outlines the unified architectural and fine-tuning configurations applied across all three candidate models.

Table 4: Unified architectural and fine-tuning configurations for the NMT models.

| Category       | Hyperparameter                       | BanglaT5                  | mBART-50                                   | NLLB-200                                   |
|----------------|--------------------------------------|---------------------------|--------------------------------------------|--------------------------------------------|
| Model Registry | HuggingFace Checkpoint               | csebuetnlp/banglat5_small | facebook/mbart-large-50-many-to-many-mmt   | facebook/nllb-200-distilled-600M           |
|                | Base Parameter Count                 | ~60M (Base) / 247M        | ~611M                                      | ~600M                                      |
|                | Src/Tgt Language Tags                | Prefixd                   | bn_IN                                      | ben_Beng                                   |
| Sequence Setup | Max Length ( $L_{src}$ , $L_{tgt}$ ) | 128 tokens                | 128 tokens                                 | 128 tokens                                 |
|                | Sequence Packing                     | Enabled                   | Disabled                                   | Disabled                                   |
| Optimization   | Optimizer                            | Paged AdamW 8-bit         | Paged AdamW 8-bit                          | Paged AdamW 8-bit                          |
|                | Learning Rate ( $\eta$ )             | $5 \times 10^{-4}$        | $5 \times 10^{-4}$                         | $5 \times 10^{-4}$                         |
|                | LR Scheduler                         | Cosine Annealing          | Cosine Annealing                           | Cosine Annealing                           |
| Batching       | Per-Device Batch Size                | 32                        | 8                                          | 8                                          |
|                | Gradient Accum. Steps                | 2                         | 8                                          | 8                                          |
|                | Effective Batch Size                 | 64                        | 64                                         | 64                                         |
| PEFT / DoRA    | Adapter Rank ( $r$ )                 | 64                        | 64                                         | 64                                         |
|                | Scaling Factor ( $\alpha$ )          | 128                       | 128                                        | 128                                        |
|                | Target Projections                   | q, v, k, o, wi, wo        | q-proj, k-proj, v-proj, out-proj, fc1, fc2 | q-proj, k-proj, v-proj, out-proj, fc1, fc2 |

## 4.5 Dataset Scaling Protocol

To formally investigate the relationship between data volume and translation generalization, we conduct an optimal dataset size study. We construct a fully parallel dataset encompassing Standard Bangla and four major dialects: Sylheti, Noakhali, Chittagong, and Mymensingh. The dataset is scaled iteratively from 500 to 4,499 parallel sentence pairs. The detailed hyperparameter matrix governing this scaling experiment is provided in Table 5.

Table 5: Hyperparameter matrix for dataset scaling experiments.

| Parameter Class          | Hyperparameter           | Configuration                                         |
|--------------------------|--------------------------|-------------------------------------------------------|
| <b>Model Config</b>      | Base Model               | csebuetnlp/banglat5                                   |
| <b>Sequence Dims</b>     | Max Sequence Length      | 128 tokens (Source/Target)                            |
| <b>PEFT Settings</b>     | DoRA Enabled             | Yes ( $r = 8, \alpha = 16$ & $r = 64, \alpha = 128$ ) |
| <b>Optimization</b>      | Optimizer                | Paged AdamW 8-bit                                     |
|                          | Learning Rate ( $\eta$ ) | $5 \times 10^{-4}$ (Cosine Decay)                     |
| <b>Batching</b>          | Effective Batch Size     | 90 (Batch size 45, Grad. Accum. 2)                    |
| <b>Training Schedule</b> | Epochs per Step          | 10 Epochs                                             |

## 4.6 Fine-Tuning Strategy

All models are fine-tuned using **Low-Rank Adaptation (LoRA)** [7], a parameter-efficient method that injects trainable low-rank decomposition matrices into each attention layer while freezing the pre-trained weights. Our experimental pipeline consists of two distinct stages:

### 4.6.1 Stage 1: Best Model Selection

In the initial stage, all three candidate models (BanglaT5, NLLB-200, mBART-50) are trained for **20 epochs** using the complete integrated dataset. To ensure a fair comparative analysis, identical LoRA configurations (Configuration B: rank  $r = 64$ ,  $\alpha = 128$ ) are applied across all models. The best-performing model is then selected based on overall evaluation metrics for the next stage.

### 4.6.2 Stage 2: Final Training

In the second stage, the selected best model (BanglaT5) is retrained for an extended period of **100 epochs** to deeply analyze learning behavior and performance saturation. Due to hardware resource constraints (the use of a Kaggle platform featuring dual T4 GPUs and 32GB RAM), our compute resources were limited, enabling us to train only the top-performing model up to 100 epochs. This extended training phase facilitates detailed error analysis and linguistic pairwise comparisons to identify dialectal proximities.

## 4.7 Multi-Directional Translation Paradigm

Unlike prior uni-directional approaches, our system is trained to handle all possible translation directions within a single unified model:

- **Dialect  $\rightarrow$  Standard Bangla:** Converting regional dialect text to SCB.
- **Standard Bangla  $\rightarrow$  Dialect:** Generating dialectal text from SCB input.
- **Dialect  $\rightarrow$  Dialect:** Direct translation between two different regional dialects without an intermediary pivot through Standard Bangla.

## 4.8 Evaluation Metrics

Model performance is assessed using four complementary automatic evaluation metrics. Given the morphological complexity of Bangla dialects, relying on a single metric is insufficient; thus, we utilize a combination of word-based, character-based, and edit-distance metrics.

### 4.8.1 BLEU (Bilingual Evaluation Understudy)

BLEU [13] measures the  $n$ -gram precision of the candidate translation against reference translations, applying a brevity penalty (BP) to discourage overly short outputs.

$$\text{BLEU} = \text{BP} \cdot \exp \left( \sum_{n=1}^N w_n \log p_n \right) \quad (1)$$



**Interpretation and Appropriateness:** Higher BLEU scores indicate greater  $n$ -gram overlap. While BLEU is the industry standard allowing for broad benchmarking, its limitation lies in its strict exact-word matching, which heavily penalizes legitimate morphological variations prevalent in Bangla dialects.

#### 4.8.2 chrF++ (Character $n$ -gram F-score)

chrF++ [15] computes the  $F$ -score based on character  $n$ -grams (typically up to 6-grams) combined with word-level bigrams.

$$\text{chrF++} = (1 + \beta^2) \frac{\text{chrP} \cdot \text{chrR}}{\beta^2 \cdot \text{chrP} + \text{chrR}} \quad (2)$$

where chrP and chrR represent character  $n$ -gram precision and recall, respectively. **Interpretation and Appropriateness:** A higher chrF++ score denotes better morphological alignment. This metric is exceptionally appropriate for highly inflected languages like Bangla, as it rewards partial word matches (e.g., matching root verbs with slightly altered dialectal suffixes) which BLEU would falsely score as zero.

#### 4.8.3 METEOR

METEOR [1] calculates the harmonic mean of unigram precision ( $P$ ) and recall ( $R$ ), placing a higher weight on recall. Crucially, it incorporates stemming and synonymy mapping.

$$\text{METEOR} = \frac{10PR}{R + 9P} \cdot (1 - \text{Penalty}) \quad (3)$$

**Interpretation and Appropriateness:** Higher scores indicate better semantic retention. The advantage of METEOR in dialectal translation is its ability to recognize synonym replacements and stem alignments, making it more aligned with human judgment regarding semantic adequacy.

#### 4.8.4 TER (Translation Edit Rate)

TER [18] measures the minimum number of human edits required to modify a generated hypothesis so that it exactly matches a reference translation.

$$\text{TER} = \frac{\text{Number of Edits}}{\text{Average Length of Reference Words}} \quad (4)$$

**Interpretation and Appropriateness:** Unlike the other metrics, a *lower* TER score indicates better performance. It is particularly useful for assessing the practical utility of the NMT system, as it provides a direct proxy for the post-editing effort required by human translators.

## 5 Experimental Results and Analysis

### 5.1 Baseline Model Comparison

Table 6 presents the overall performance of the three NMT models after fine-tuning on the full poly-dialectal corpus. BanglaT5 consistently outperforms the two larger multilingual models despite having only 247M parameters—less than half the size of NLLB-200 (615M) and mBART-50 (611M).

Table 6: Overall performance comparison of the three NMT models on the poly-dialectal test set (18,062 translation pairs across all dialect directions). Best results are shown in **bold**.

| Model             | Params | BLEU $\uparrow$ | chrF++ $\uparrow$ | METEOR $\uparrow$ | TER $\downarrow$ |
|-------------------|--------|-----------------|-------------------|-------------------|------------------|
| BanglaT5 (10 ep)  | 247M   | 23.22           | 51.20             | 41.41             | 56.85            |
| NLLB-200          | 615M   | 14.76           | 42.62             | 33.56             | 64.53            |
| mBART-50          | 611M   | 8.96            | 32.94             | 23.82             | 77.10            |
| BanglaT5 (100 ep) | 247M   | <b>29.26</b>    | <b>57.26</b>      | <b>49.68</b>      | <b>50.59</b>     |

The extended 100-epoch BanglaT5 training yields a 26% relative improvement in BLEU (from 23.22 to 29.26) over the 10-epoch baseline, demonstrating that dialect translation benefits substantially from prolonged fine-tuning. Figure 4 visualizes this comparison across all four metrics.

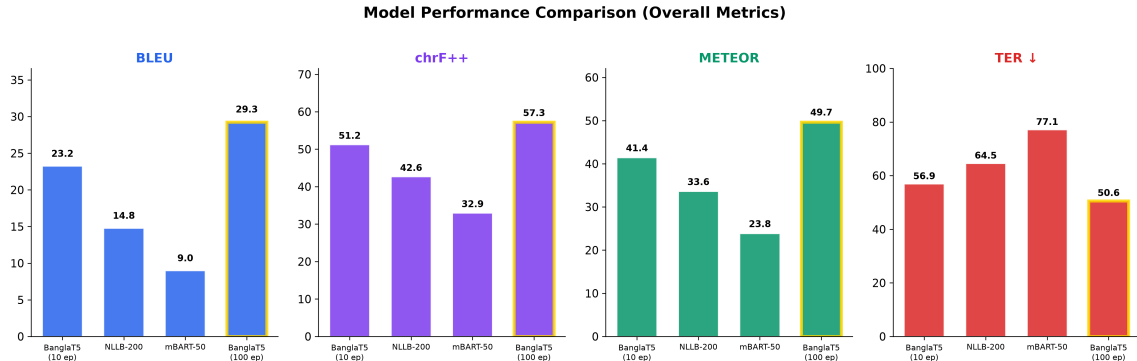


Figure 4: Comparative evaluation of four model configurations across BLEU, chrF++, METEOR, and TER. Gold-bordered bars indicate the best-performing configuration.

### 5.2 Cross-Dialectal Translation Analysis

Figure 5 presents a heatmap of BLEU scores for all source-target dialect pairs under the best model (BanglaT5, 100 epochs). Several key findings emerge:

1. **Morphologically proximate dialects translate better:** Mymensingh  $\rightarrow$  Standard Bangla achieves the highest BLEU score of 55.0, consistent with Mymensingh’s closer lexical similarity to SCB. Conversely, Rajshahi and Rangpur—dialects with distinct phonological systems—yield BLEU scores below 12 in most directions.

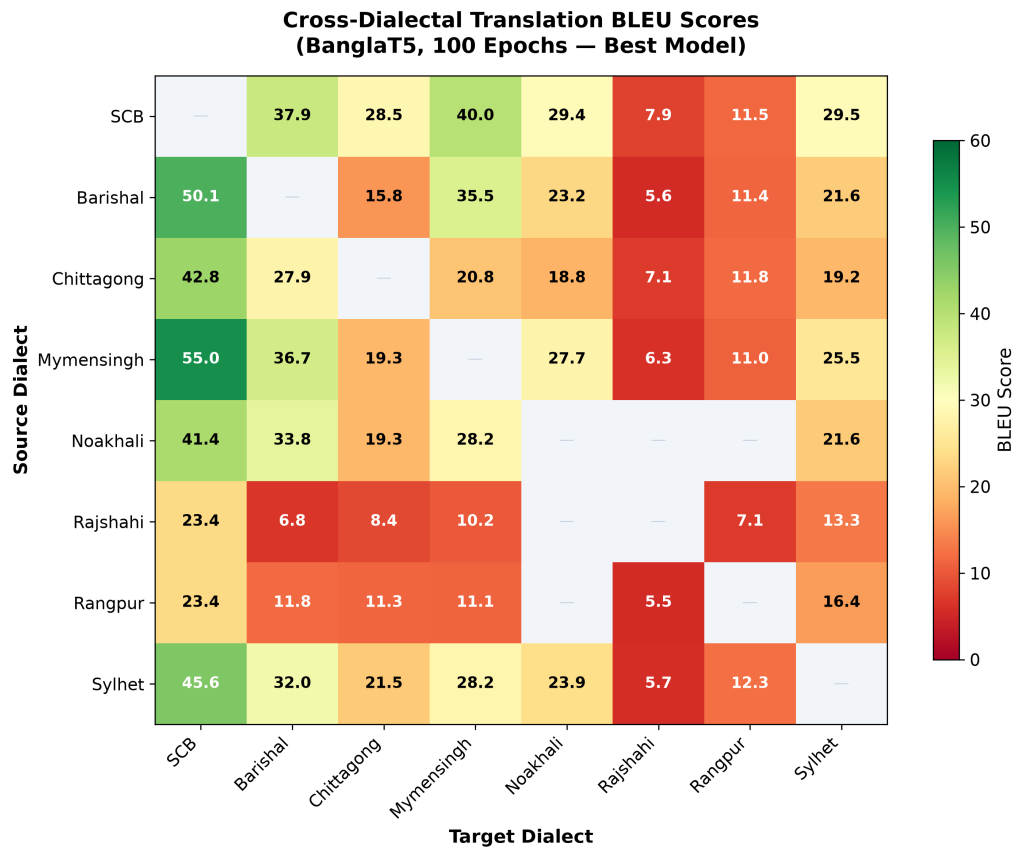


Figure 5: Cross-dialectal BLEU score heatmap (BanglaT5, 100 epochs). Darker green indicates higher translation quality. Rajshahi and Rangpur (bottom rows) exhibit uniformly low scores due to limited training data.

2. **Translation directionality is asymmetric:** As shown in Figure 6, translating *from* a dialect to Standard Bangla is consistently easier than the reverse. For example, Barisali  $\rightarrow$  SCB achieves 50.1 BLEU while SCB  $\rightarrow$  Barisali achieves only 37.9 BLEU, a gap of 12.2 points. This asymmetry reflects the decoder’s greater difficulty in generating dialectal morphological forms absent from its pre-training distribution.
3. **Data volume directly correlates with performance:** Dialects with fewer than 1,000 training samples (Rajshahi: 891, Rangpur: 1,155) perform drastically worse than those with 5,000+ samples, underscoring the critical role of corpus volume in NMT quality.

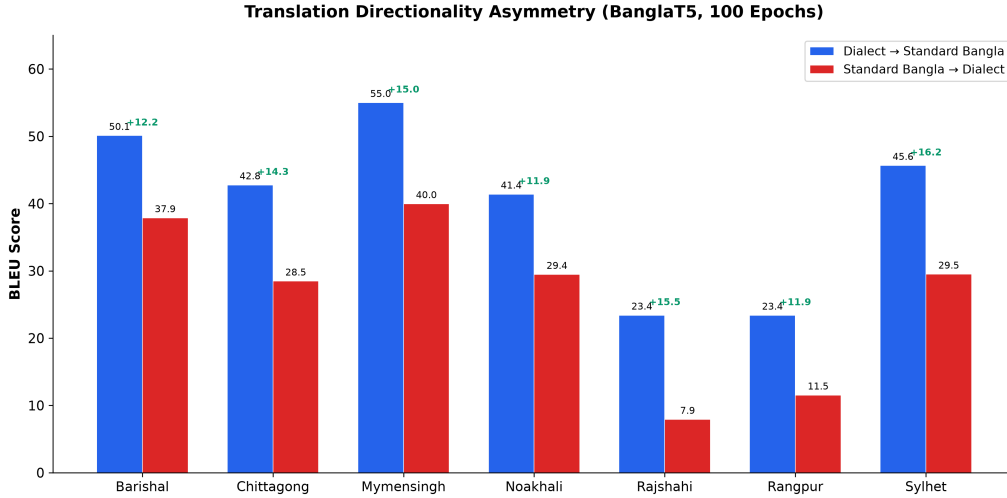


Figure 6: Translation directionality asymmetry: Dialect $\rightarrow$ SCB consistently outperforms SCB $\rightarrow$ Dialect, with green labels indicating the BLEU gap.

### 5.3 Dataset Scaling Analysis

To quantify the relationship between training data volume and translation quality, we conduct a controlled scaling experiment using BanglaT5 with LoRA fine-tuning. Training data is incrementally scaled from 500 to 4,499 sentence pairs per dialect combination, with results measured on a fixed test set of 10,000 pairs.

As shown in Figure 7, BLEU improves from 12.09 to 25.74 (a 113% relative increase) as data scales from 500 to 4,499 under the  $r=8$  configuration. The higher-capacity  $r=64$  configuration yields slightly better results (27.55 BLEU at 4,499 samples) but shows diminishing marginal returns beyond approximately 3,000 samples. This finding validates the central premise of this thesis: *larger, more diverse corpora are essential for effective poly-dialectal NMT*.

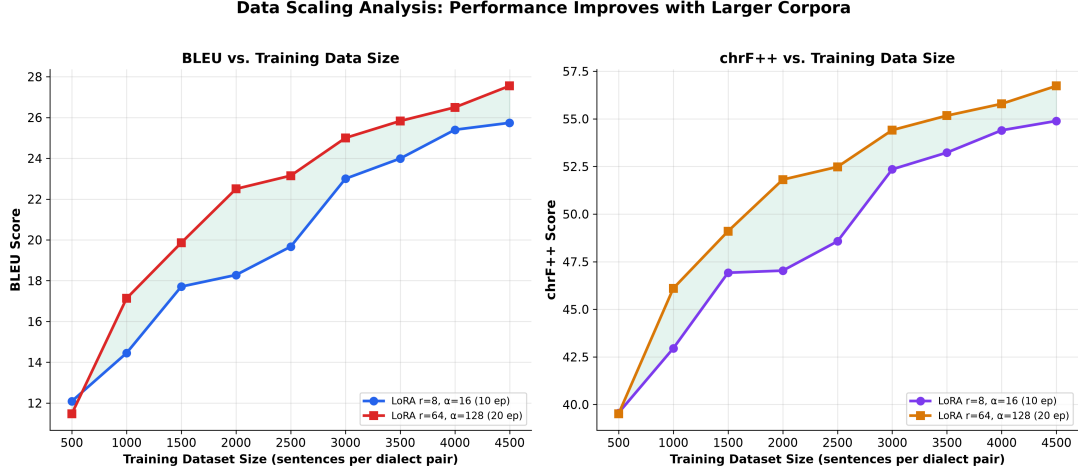


Figure 7: BLEU and chrF++ scaling curves as training data increases from 500 to 4,499 per dialect pair. Both LoRA configurations show consistent improvement with larger data, but with diminishing returns beyond 3,000 samples.

## 6 Limitations and Future Work

### 6.1 Current Limitations

1. **Persistent Class Imbalance:** Despite integrating seven source corpora, the dataset exhibits a 21:1 imbalance ratio between the most represented dialect (Standard Bangla: 14,535 entries) and the least represented (Kishoreganj, Narail, Narsingdi, Tangail:  $\sim 502$  each). This imbalance limits the model’s generalization capacity for underrepresented dialects, as evidenced by the low BLEU scores for Rajshahi (23.4) and Rangpur (23.4) even under the best model.
2. **Computational Cost:** Fine-tuning even moderately sized models (247M–615M parameters) on the full multi-directional translation matrix requires substantial GPU memory and training time. The 100-epoch BanglaT5 training required approximately 851 seconds, while mBART-50 required 13,889 seconds—a  $16\times$  disparity highlighting the efficiency advantage of language-specific models.
3. **Text-Only Modality:** The current system operates exclusively on text input, neglecting the phonological richness and prosodic variation characteristic of spoken Bangla dialects. A speech-to-text pipeline would be necessary for real-world deployment in voice-based applications.
4. **Absence of Human Evaluation:** While automatic metrics (BLEU [13], chrF++ [15], METEOR [1], TER [18]) provide objective and reproducible quality assessments, they cannot fully capture translation fluency, dialectal authenticity, and semantic adequacy as perceived by native speakers.

### 6.2 Future Directions

1. **Corpus Expansion:** Systematic data collection for underrepresented dialects through community-driven annotation campaigns and web scraping of social media platforms.

2. **Multimodal Integration:** Extending the framework to incorporate speech-to-text and text-to-speech modules, enabling end-to-end dialectal speech translation.
3. **Advanced Training Strategies:** Investigating curriculum learning, data augmentation through back-translation, and multi-task training to improve performance on low-resource dialect pairs.
4. **Human Evaluation Protocol:** Designing and conducting a formal human evaluation study with native dialect speakers to assess translation adequacy, fluency, and dialectal authenticity.

## 7 Conclusion

This thesis has presented a comprehensive framework for poly-dialectal neural machine translation in Bangla, addressing the critical gap between the rich dialectal diversity of the language and the limited scope of existing NLP tools. The principal achievements of this work are threefold.

First, we constructed the **largest multi-dialectal parallel corpus** for Bangla machine translation to date, integrating seven existing datasets into a unified resource comprising 14,552 aligned entries across 12 regional dialect columns, totaling 51,531 parallel sentence pairs. This corpus introduces coverage for four previously unaddressed dialects (Kishoreganj, Narail, Narsingdi, and Tangail) and provides up to  $2.76\times$  more data per dialect than the best existing competitor.

Second, our systematic **model evaluation** revealed that BanglaT5—a language-specific model with only 247M parameters—decisively outperforms the larger multilingual models NLLB-200 (615M) and mBART-50 (611M), achieving a best overall BLEU of 29.26 and chrF++ of 57.26 after 100 epochs of LoRA fine-tuning. This finding underscores the importance of language-specific pre-training for low-resource intra-language translation tasks.

Third, our **cross-dialectal analysis** uncovered fundamental patterns in dialect translatability: (a) morphologically proximate dialects yield higher BLEU scores (Mymensingh→SCB: 55.0), (b) dialect-to-standard translation is consistently easier than the reverse by 10–15 BLEU points, and (c) translation quality scales log-linearly with training data volume, with diminishing returns observed beyond 3,000 samples per dialect pair.

By enabling direct, multi-directional translation without a pivot through Standard Bangla, this system represents a meaningful step toward inclusive, dialect-aware NLP infrastructure for the Bangla-speaking population. The corpus, models, and evaluation framework established in this thesis provide a robust foundation for future research in low-resource dialectal machine translation.

## References

- [1] Satanjeev Banerjee and Alon Lavie. Meteor: An automatic metric for mt evaluation with improved correlation with human judgments. In *Proceedings of the acl workshop on intrinsic and extrinsic evaluation measures for machine translation and/or summarization*, pages 65–72, 2005.
- [2] Abhik Bhattacharjee, Tahmid Hasan, Wasi Uddin Ahmad, and Rifat Shahriyar. BanglaNLG and BanglaT5: Benchmarks and resources for evaluating low-resource natural language generation in Bangla. In *Findings of the Association for Computational Linguistics: EACL 2023*, pages 726–735, 2023.
- [3] M. Bhuiyan et al. Bridging bangla dialects and romanized forms through machine translation. *Preprint*, 2025.
- [4] T. Chowdhury et al. A dataset for conversion from chittagonian dialect to standard bangla. *Data in Brief*, 59:111200, 2025.
- [5] I. Faria et al. A large-scale multilingual benchmark dataset for translation of bangla regional dialects. *arXiv preprint arXiv:2503.00004*, 2025.

- [6] S. Fayaz et al. A bangla regional based hate speech detection dataset. *arXiv preprint arXiv:2502.00003*, 2025.
- [7] Edward J Hu, Yelong Shen, Phil Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. Lora: Low-rank adaptation of large language models. In *International Conference on Learning Representations*, 2022.
- [8] M. Jawad et al. Benchmarking llms on bangla dialect translation and sentiment analysis. In *Proceedings of the Workshop on Bangla Language Processing (BLP)*, pages 55–65, 2025.
- [9] A. Kundu et al. A comprehensive corpus for sentiment analysis in bangla regional languages. *arXiv preprint arXiv:2601.00002*, 2026.
- [10] Mike Lewis, Yinhan Liu, Naman Goyal, Marjan Ghazvininejad, Abdelrahman Mohamed, Omer Levy, Veselin Stoyanov, and Luke Zettlemoyer. Bart: Denoising sequence-to-sequence pre-training for natural language generation, translation, and comprehension. *arXiv preprint arXiv:1910.13461*, 2019.
- [11] M. Mahi et al. A merged and imbalanced text dataset for bengali regional dialect analysis. *Data in Brief*, 58:111000, 2025.
- [12] S. Mahjabin et al. Human-llm benchmarks for bangla dialect translation. In *Proceedings of the Workshop on Bangla Language Processing (BLP)*, pages 45–54, 2025.
- [13] Kishore Papineni, Salim Roukos, Todd Ward, and Wei-Jing Zhu. Bleu: a method for automatic evaluation of machine translation. In *Proceedings of the 40th annual meeting of the Association for Computational Linguistics*, pages 311–318, 2002.
- [14] S. Paul et al. A benchmark dataset for bangla regional named entity recognition. *arXiv preprint arXiv:2501.00001*, 2025.
- [15] Maja Popović. chrF: character n-gram f-score for automatic mt evaluation. In *Proceedings of the Tenth Workshop on Statistical Machine Translation*, pages 392–395, 2015.
- [16] Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, and Peter J Liu. Exploring the limits of transfer learning with a unified text-to-text transformer. *Journal of Machine Learning Research*, 21(140):1–67, 2020.
- [17] Rico Sennrich, Barry Haddow, and Alexandra Birch. Neural machine translation of rare words with subword units. In *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1715–1725, 2016.
- [18] Matthew Snover, Bonnie Dorr, Richard Schwartz, Linnea Micciulla, and John Makhoul. A study of translation edit rate with targeted human annotation. In *Proceedings of the 7th Conference of the Association for Machine Translation in the Americas: Technical Papers*, pages 223–231, 2006.
- [19] N. Sultana et al. A comprehensive dataset for automated conversion of bangla regional dialects. *Data in Brief*, 60:111500, 2025.



- [20] R. Sultana et al. Resource creation and evaluation for bangla regional text classification. *Indonesian Journal of Electrical Engineering and Computer Science*, 34(2):1120–1129, 2024.
- [21] Yuqing Tang, Chau Tran, Xian Li, Peng-Jen Chen, Naman Goyal, Vishrav Chaudhary, Jiatao Gu, and Angela Fan. Multilingual translation with extensible multilingual pretraining and finetuning. *arXiv preprint arXiv:2008.00490*, 2020.
- [22] NLLB Team, Marta R. Costa-jussà, et al. No language left behind: Scaling human-centered machine translation. *arXiv preprint arXiv:2207.04672*, 2022.
- [23] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. *Advances in neural information processing systems*, 30, 2017.
- [24] Yonghui Wu, Mike Schuster, Zhifeng Chen, Quoc V Le, Mohammad Norouzi, Wolfgang Macherey, Maxim Krikun, Yuan Cao, Qin Gao, Klaus Macherey, et al. Google’s neural machine translation system: Bridging the gap between human and machine translation. *arXiv preprint arXiv:1609.08144*, 2016.